

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:CSA0716

COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _Ysiddartha /192425047_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and wellstructured explanation	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Answer

1. Missing OSI Layers

Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Missing Layers

- Session Layer
- Data Link Layer

Why they are essential

Session Layer

- Establishes communication between devices.
- Maintains and terminates sessions.
- Provides synchronization during data exchange.

Data Link Layer

- Converts packets into frames.
- Uses MAC addresses for local delivery.
- Detects transmission errors and controls access to the physical medium.

Conclusion

The Session layer ensures organized communication, while the Data Link layer ensures accurate local transmission, making communication more reliable.

2. Concept Map for Data Units

Transport

Segment

Network

Packet

Data Link

Frame

Physical

Bits

Analysis

- Segment: Adds port numbers for process communication.
- Packet: Adds logical IP addresses for routing.
- Frame: Adds MAC addresses and error-checking information.
- Bits: Converts data into electrical, optical, or radio signals.

Conclusion

Each OSI layer encapsulates the data with additional control information, enabling successful transmission across networks.

3. MAC Address and IP Address Relationship

Concept Map

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Explanation

MAC Address

- Identifies a device within the same local network.
- Used by switches for local communication.

IP Address

- Identifies a device across different networks.
- Used by routers for end-to-end delivery.

How they work together

- The IP address moves the packet from source network to destination network.
- The MAC address delivers the frame to the correct device on the local network.

Conclusion

IP provides global routing, while MAC provides local delivery, ensuring complete communication from sender to receiver.

4. Error Control Interaction

Concept Map

Error Control

└─ Data Link Layer → Error Detection

└── Transport Layer → Error Correction

Analysis

Data Link Layer

- Detects errors using CRC or checksum.
- Discards corrupted frames.

Transport Layer

- Detects missing or damaged segments.
- Requests retransmission (TCP).
- Ensures correct order and complete delivery.

Interaction

- The Data Link layer handles errors on a single link.
- The Transport layer handles errors across the entire network path.

Conclusion

Combining error detection and error correction significantly improves reliability and reduces data loss.

5. Concept Map for Web Browsing

Application

HTTP / HTTPS

Presentation

SSL / TLS

Session

Session control

Transport

TCP

Network

IP routing

Data Link

Ethernet / Wi-Fi

Physical

Signals / cable / radio

Effect of Failure

Layer Failure	Effect
Physical	No signal; webpage cannot load.
Data Link	Frames cannot reach the router.
Network	IP routing fails; destination unreachable.
Transport	TCP connection cannot be established.
Application	Browser cannot request or display the webpage.

Conclusion

Web browsing depends on all OSI layers. Failure in one layer affects the layers above it, causing the entire communication process to fail.

6. Troubleshooting Using the Concept Map

Given

Physical ✓

Data Link ✓

Network ✗

Transport ✗

Application ✗

Analysis

- Physical ✓ → Cable/Wi-Fi connection is working.
- Data Link ✓ → Local communication is successful.
- Network ✗ → Problem occurs at the IP layer.

Possible Causes

- Incorrect IP address.
- Wrong subnet mask.
- Missing default gateway.
- Router failure.
- Routing table problem.

Why Transport and Application Fail

These layers depend on the Network layer. If packets cannot be routed, TCP/UDP communication and applications cannot function.

How the Concept Map Helps

- Physical → Check cable, power, Wi-Fi.
- Data Link → Check MAC address and switch connectivity.
- Network → Check IP configuration and routing.
- Transport → Test TCP/UDP ports.
- Application → Verify browser, DNS, or server service.

Conclusion

The concept map helps identify that the root cause is the Network layer, making troubleshooting.